

Table 9: **Transferability of greedy search IC prompt to other LLMs.** We use the 1st-turn score on MT-Bench and the length-controlled (LC) win rates on AlpacaEval (AE) 2.0 when using various IC prompts on different base models.

Model	MT-B (1st)	AE 2.0
Base model: Llama-2-7B-80k		
URIAL (3 examples)	5.19	1.81
URIAL + greedy search (1 ex.)	5.56	1.50
URIAL + greedy search (2 ex.)	5.57	1.84
URIAL + greedy search (3 ex.)	5.10	2.91
Base model: Mixtral-8x22B-v0.1-4bit		
URIAL (3 examples)	8.28	14.77
URIAL + greedy search (1 ex.)	8.36	15.74
URIAL + greedy search (2 ex.)	7.79	13.20
URIAL + greedy search (3 ex.)	8.58	14.68

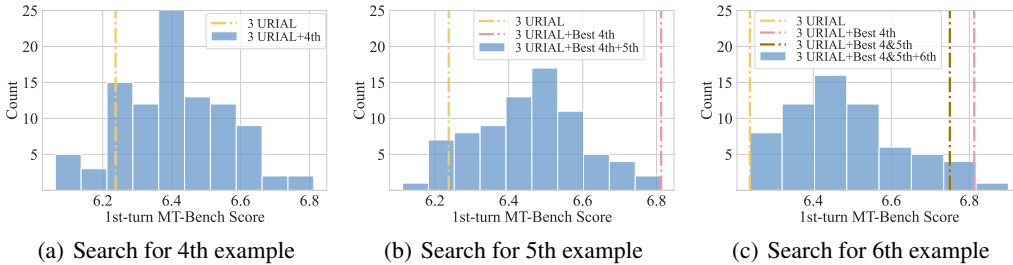


Figure 12: **The distribution of the 1st-turn MT-Bench score (GPT-4-Turbo as judge) on Mistral-7B-v0.2 obtained by adding multiple instructions from SkillMix (Kaur et al., 2024) as a 4th (a); 5th (b); 6th (c) demonstration to URIAL.** The dashed lines of various colors refer to the 1st-turn MT-Bench score of the obtained searching results. A majority of the 4th examples have positive contributions to the model’s instruction-following performance, but the improvement quickly diminishes when running the greedy search for more optimal demonstrations.

KL-divergence value rapidly drops to approximately 0. However, the KL-divergence between the IFT models and the base model does not converge to 0 and has larger KL-divergence values on average. Surprisingly, IFT on only 3 examples is already sufficient to drive the fin-tuned model far away from the base model as measured by KL-divergence compared to in-context alignment. In particular, the difference in the probability distribution of next token prediction becomes more substantial as the base model are instruction fine-tuned on more data.

D TRANSFERABILITY OF IN-CONTEXT PROMPTS

In Table 9, we report the performance of applying the in-context examples found by greedy search (added to URIAL) on Mistral-7B-v0.2 and Evol-Instruct-70k dataset (see Table. 6) to Llama-2-7B-80k (Fu et al., 2024) and Mixtral-8x22B-v0.1-4bit (Jiang et al., 2024). Adding the new examples does not provide a consistent improvement: while it can increase the MT-Bench score (+0.47 on Llama-2-7B-80k, +0.30 on Mixtral-8x22B-v0.1-4bit), it can also, in some cases, give worse results than the original URIAL. Similarly, it yields mixed results when measured by win rate on AlpacaEval 2.0, which was not optimized by the greedy search. This analysis suggests that the most effective ICL demonstrations might vary across base LLMs, potentially because of differences in their pre-training. This could further explain why URIAL underperforms on multiple models in Table 1, especially on those which have become available only recently and were not used for selecting the URIAL examples.

Table 10: **Scaling experiment results (length-controlled win-rates) on AlpacaEval 2.0.** We put different amounts of random examples drawn from SkillMix to the context as demonstrations. We conduct experiments on two base models: Mistral-7B-v0.2 and Meta-Llama-3.1-8B. The results are computed over 3 different runs.

	AlpacaEval 2.0 LC Win-rate (%)	
	Mistral-7B-v0.2	Meta-Llama-3.1-8B
0 example	2.92 ± 0.0	2.99 ± 0.0
1 example	11.98 ± 0.7	8.66 ± 2.7
3 examples	12.50 ± 1.3	11.56 ± 0.6
10 examples	14.86 ± 0.4	15.11 ± 1.9
20 examples	13.20 ± 0.7	15.18 ± 0.8
30 examples	12.45 ± 0.1	13.90 ± 0.2
40 examples	12.51 ± 1.3	14.95 ± 0.7
50 examples	13.80 ± 0.7	15.49 ± 0.8

E ADDITIONAL EXPERIMENTS

E.1 SCALING EXPERIMENTS ON ALPACAEVAL 2.0

This section provides additional results of scaling experiments conducted on the AlpacaEval Suite, which contains 805 single-turn examples, an order of magnitude more than MT-Bench. Based on the SkillMix dataset, we construct the set of in-context demonstrations by sampling different amounts of random examples, ranging from 0 to 50 examples. The results shown in Table 10 confirm our findings, reported in Sec. 3, that scaling the number of ICL examples is not sufficient to consistently improve the instruction following performance.

E.2 GREEDY SEARCH ON MISTRAL-7B-v0.2

In this section, we show the distribution of the resulting scores (with GPT-4-Turbo as the judge) for each step of the greedy search on SkillMix-4k dataset when using Mistral-7B-v0.2 as the base model. Similar to the finding on Llama-3.1-8B model (see Sec. 3.2), the majority of demonstrations increase the alignment performance, as measured by 1st-turn MT-Bench score, when added as the fourth example on top of URIAL, and few progress is made with more demonstrations. We show more details of greedy search results on Mistral-7B-v0.2 in Fig. 12.

E.3 MORE EXPERIMENTS ON DECODING SCHEMES

We show more results of decoding schemes under different settings in this section, as supplementary to the main results in Sec. 2.2. Specifically, in Fig. 13, we turn off the repetition penalty (i.e., repetition penalty = 1.0) and show the 1st-turn MT-Bench scores on Mistral-7B-v0.2 when varying the values of temperature and top-p. Moreover, we repeat the same experiment procedure but on a different base model, Llama-3-8B, and show heatmaps in Fig. 14.

E.4 A BREAKDOWN EXAMINATION OF URIAL-ALIGNED MODELS

The test set of MT-Bench (Zheng et al., 2023) is comprised of high-quality questions that belong to 8 common categories: coding, math, reasoning, extraction, roleplay, writing, humanities/social science, and STEM. In Table 11, we present the per-category performance on MT-Bench for each model in order to have a more detailed comparison between the URIAL-aligned models and the corresponding instruct models. The results suggest fine-tuning is almost always better than URIAL (ICL) and only underperforms in a minority of cases on particular base models, such as Mistral-7B-v0.1 and Llama-2-70B.

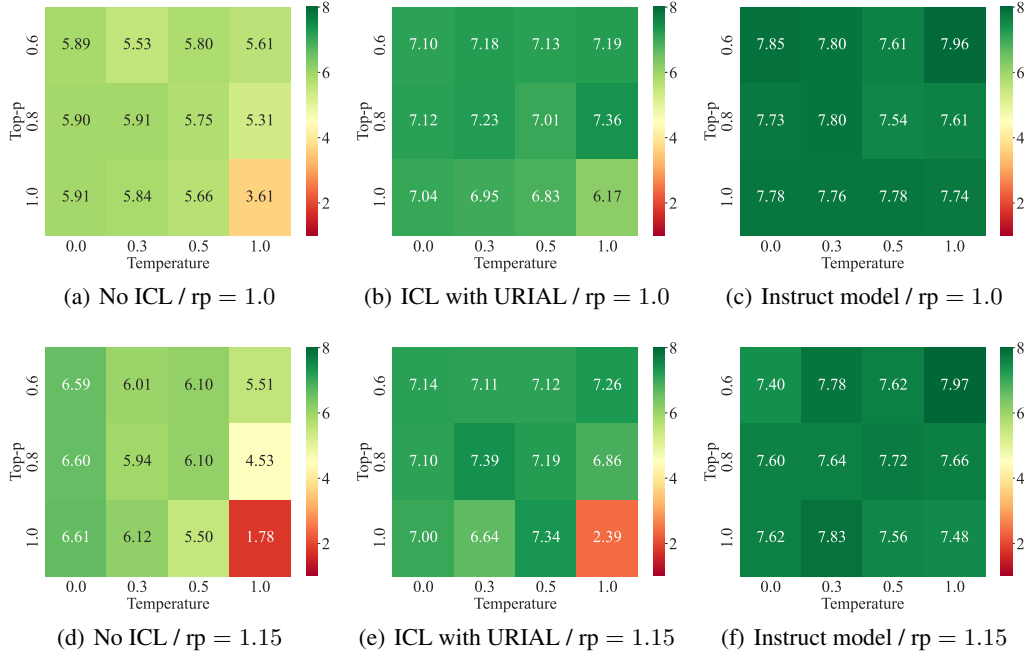


Figure 13: **The 1st-turn MT-Bench scores of model generations with and without ICL across different decoding schemes.** We mainly consider two hyper-parameters: temperature and top-p. The heatmaps show the answering quality of the base model with and without URIAL in the context is sensitive to the setups of decoding schemes. Surprisingly, with proper decoding parameters, the base model alone is already capable of following instructions.

Table 11: **A breakdown examination of URIAL-aligned models and corresponding instruct models across different categories on MT-Bench.** * denotes the results taken from the URIAL GitHub repository.

Model	coding	math	reasoning	extraction	humanities	roleplay	stem	writing	Average
Llama-2-7B + URIAL *	1.65	1.60	3.45	3.40	8.08	7.48	6.80	6.20	4.83
Llama-2-7B-Instruct	2.95	2.40	5.20	6.33	9.58	7.83	8.88	9.05	6.53
Llama-2-70B + URIAL *	4.15	3.60	6.10	7.70	9.75	7.33	8.75	9.50	7.11
Llama-2-70B-Instruct	3.75	4.10	5.95	7.40	9.85	7.90	9.13	9.50	7.20
Llama-3-8B + URIAL *	4.15	2.60	3.50	5.25	8.90	7.30	8.15	6.13	5.75
Llama-3-8B-Instruct	5.95	5.05	6.15	9.16	9.90	9.05	8.95	8.70	7.86
Llama-3-70B + URIAL *	4.35	3.80	4.95	6.20	8.00	7.25	8.55	8.10	6.40
Llama-3-70B-Instruct	7.85	7.35	6.25	9.75	10.00	9.30	9.60	9.80	8.74
Llama-3.1-8B + URIAL *	4.35	3.25	3.95	5.95	9.00	6.95	8.00	7.60	6.13
Llama-3.1-8B-Instruct	6.40	6.50	5.70	8.78	9.80	9.00	8.60	9.20	8.00
Mistral-7B-v0.1 + URIAL *	4.60	3.40	4.90	7.75	9.08	7.65	8.28	7.75	6.67
Mistral-7B-Instruct-v0.1	4.35	3.95	6.30	6.75	9.45	7.45	7.70	8.85	6.85
Mistral-7B-v0.2 + URIAL *	3.80	3.35	4.50	7.45	8.95	6.70	7.43	7.98	6.27
Mistral-7B-Instruct-v0.2	5.45	3.40	6.50	8.50	9.90	8.65	9.30	9.40	7.64
Mixtral-8x22B-v0.1-4bit + URIAL	5.25	5.85	6.00	9.20	9.80	8.65	8.30	8.60	7.71
Mixtral-8x22B-Instruct-v0.1-4bit	7.10	7.03	7.00	9.10	9.65	8.90	9.65	9.70	8.52
GPT-4-Base + URIAL	5.55	5.98	6.90	8.20	5.93	6.90	6.45	6.10	6.50
GPT-4 (March 2023) *	8.55	6.80	9.00	9.38	9.95	8.90	9.70	9.65	8.99

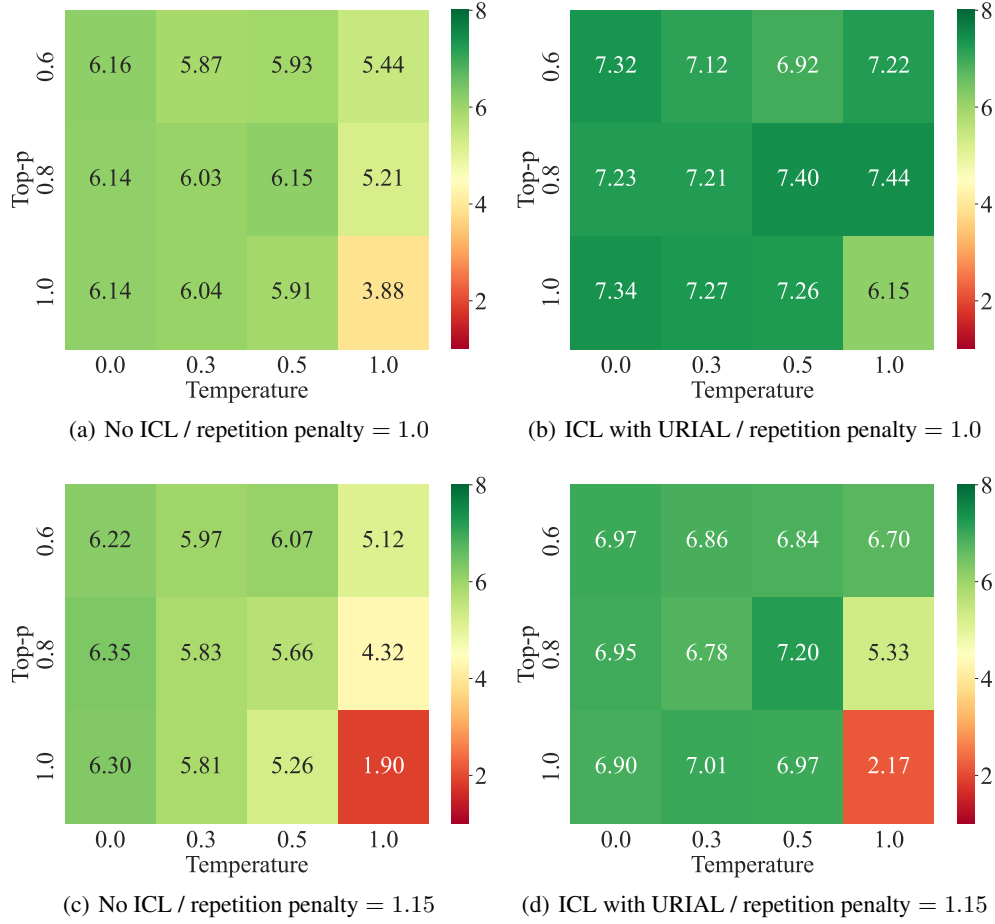


Figure 14: **The 1st-turn MT-Bench scores of Llama-3.1-8B generations with and without URIAL in the context across different decoding schemes.** We mainly consider two hyper-parameters: temperature and top-p.